SINF1 Project – 2025/2026

1 Introduction

The objective of this practical work is to develop a Web-based Information System (IS) related to an academic cultural event. All information must be stored persistently in a relational database.

2 Description

Context

The intended Information System is designed to manage events, spaces and participants associated with Queima das Fitas do Porto.

This academic festival includes:

- Academic ceremonies (Serenata, Cortejo Académico, Missa da Bênção das Pastas, etc.)
- Concert nights
- Cultural activities
- Faculty tents (barracas)
- Artists and performances
- Student participation

The system must allow the management and visualisation of all relevant information related to the festival.

Global Requirements

The system must allow:

- Registration and authentication of users.
- Different access roles (Administrator and Student).
- Management of multiple types of events.
- Management of faculty tents associated with different faculties.
- Association of artists with concert events.
- Classification and ordering of events according to:
    - Date
    - Popularity
    - Rating
- Rating of events and tents (1–5 scale).
- Alerts for upcoming events (e.g., events occurring within 48 hours).
- Import of relevant data (e.g., events or artists) from CSV files.
- Export of selected information to CSV format.

Other requirements:

- All registered users must be able to view the festival programme.
- Only authorised users (administrators) may create, modify or remove core entities.
- The system must be intuitive and suitable for end users without technical knowledge.
- The system must be as interactive as possible. As an example, a given item that appears on a given page must always have an associated link to the information page about the item.

Other information that is relevant to the operation of the proposed system may be included. The addition of features that prove important to the system will be valued.

3 MINIMAL SYSTEM REQUIREMENTS

USER MANAGEMENT

The system must support:

- User registration
- Login and logout
- User profile page (edit personal information)

Each user must have one of the following roles:

- Administrator
- Student

Students may:

- View events, tents and artists
- Create a personal agenda
- Rate events and tents

Administrators may:

- Perform full CRUD operations over all entities

EVENTS

The system must support different event types, such as:

- Academic ceremony
- Concert
- Cultural activity

Each event must include at least:

- Name
- Description
- Date and time
- Location
- Type
- Associated tent (if applicable)
- Associated artist(s) (for concerts)

The system must alert users when an event date is approaching.

FACULTIES

Each faculty must include at least:

- Name
- Acronym
- Description
- Representative colour (optional attribute)

A faculty may be associated with one or more tents.

ARTISTS

Each artist must include:

- Name
- Musical genre
- Country
- Short biography

An artist may be associated with multiple events.

FACULTY TENTS

Each tent must include:

- Name
- Associated faculty
- Location within the recinto
- Opening and closing hours
- Description

Students may rate tents after visiting.

PERSONAL AGENDA

Students must be able to:

- Add events to a personal agenda
- View upcoming selected events
- Remove events from their agenda

RATINGS

Students may rate:

- Events
- Faculty tents

Ratings must use a scale from 1 to 5.

The system must calculate and display the average rating.

4 DATABASE REQUIREMENTS

The system must be supported by a MySQL relational database.

The following entities must be considered (minimum):

- User
- Role
- Faculty
- Tent
- Event
- Artist
- Rating
- PersonalAgenda

Relationships must be properly modelled (including many-to-many relationships where applicable).

The following artefacts are mandatory:

- Conceptual model (ER diagram).
- Logical model.
- Physical model (SQL script including table creation and test data).

5 TECHNOLOGIES

The project must use:

- PHP (backend logic and CRUD operations)
- MySQL (DBMS)
- HTML5
- CSS3
- JavaScript / DOM manipulation
- Git and Github for version control
  - The repository name must follow the pattern SINF1_2025-2D**X**_G**YY**, where **X** is the class id **YY** is the group number (example: SINF1_2025-2D**A**_G**01** is the repository of group G01 of class 2DA).
  - Teachers should be given access to the repository

Frequent commits with meaningful messages are mandatory.

6 GRADE IMPROVEMENT CHALLENGES

The following features are optional but may contribute to grade improvement:

- Search and filtering system (by date, faculty, type, rating).
- Dashboard with statistics (e.g., most popular event, highest-rated tent).
- Role-based middleware logic with session control.
- Image upload for events or artists.
- Dynamic countdown timer for upcoming events.
- Comment system associated with ratings.
- Data validation with both client-side and server-side checks.
- Responsive design.

7 DELIVERABLES

Submission Deadline: 20/05/2026

Submission via Moodle must include:

- Complete Web application (PHP, HTML, CSS, JavaScript files).
- SQL script for database creation including test data.
- ER diagram (conceptual model).
- Self- and peer-assessment document.

All files must be compressed into a single .zip file.

8 ASSESSMENT

The final evaluation will consider:

- Functional completeness.
- Correct modelling of relationships.
- Quality of database design.
- Code organisation and readability.
- Proper use of version control.
- Interface usability.
- Implementation of optional advanced features.
- Oral presentation and technical justification.

Late submissions within 24 hours incur a 10% penalty.

Submissions after this period will not be accepted.